

WW2 WINGS



NORMANDY · 1944

*A pilot's career in three mini-games.
A tribute to Cinemaware's Wings (1990).*

A career, one sortie at a time

You are a replacement pilot posted to the 406th Fighter Squadron in the first days of June 1944. You fly the P-47 Thunderbolt, and the invasion of France is about to begin.

WW2 Wings follows the shape of Cinemaware's 1990 Amiga classic *Wings*, moved from the Great War to the Second. The story is told through your pilot's diary. Between entries you fly, and every sortie is one of three different mini-games.

- **Air combat** in full 3D, seen from behind your aircraft.
- **Bombing** from directly overhead.
- **Ground attack** at tree-top height, seen from an isometric camera.

Around the flying sits the career: a pilot you create, aptitudes that grow with experience, promotions and medals, and the real chance that he does not come home. When that happens a replacement takes his bunk and the war goes on.



The main menu. The flight of Thunderbolts behind it is rendered live.

3

MINI-GAMES

18

MISSION VARIANTS

18

CAMPAIGN MISSIONS

4

PILOT APTITUDES

0

ART OR SOUND FILES

Air combat



Escort duty: a box of B-26 Marauders with the Thunderbolts above them.



Contacts are bracketed with their range; friendly aircraft are marked in blue. Distant aircraft are drawn larger than life.

Dogfights against Bf 109s and Fw 190s, flown from a chase camera with your wingmen beside you.

- Roll toward the enemy, then pull. Banked wings carry the nose around, and the aircraft eases back to level when you let go of the stick.
- A **lead marker** shows where to shoot; brackets, range and edge arrows keep track of every bandit.
- Damaged aircraft trail smoke, handle worse and finally go down burning.
- Ammunition is limited. Run dry and you break off for home.
- **Six variants:** fighter sweep, escorting B-26 Marauders, intercepting Ju 88s with tail gunners, chasing V-1 flying bombs, being bounced from behind, and a duel with an ace.

IN THE ORIGINAL

Wings showed its dogfights from behind the pilot's head. The idea is kept; the view is pulled back to a modern chase camera.

Bombing



A stick of bombs walks through the rail yard at Amiens.



A goods train caught on the main line on the way in.

A run over the target from directly above, with a limited bomb load and flak coming up to meet you.

- Bombs take time to fall. A **ring on the ground** shows where one released now will land.
- Rolling stock, warehouses and fuel dumps burn, and fuel sets off whatever stands next to it.
- Flak is aimed at where you are going to be. Keep moving.
- Miss the objective and you can come around for one more pass.
- **Six variants:** rail yard, bridge, a moving armoured column, a harbour full of barges, a V-1 launch site under heavy flak, and an airfield.

IN THE ORIGINAL

The top-down bombing run is the closest of the three to the 1990 game. The impact marker and the second pass are additions.

Ground attack



The coastal battery: bunkers in the dunes, wrecked landing craft in the surf.



A flak position beside the Caen road, about to be hit.

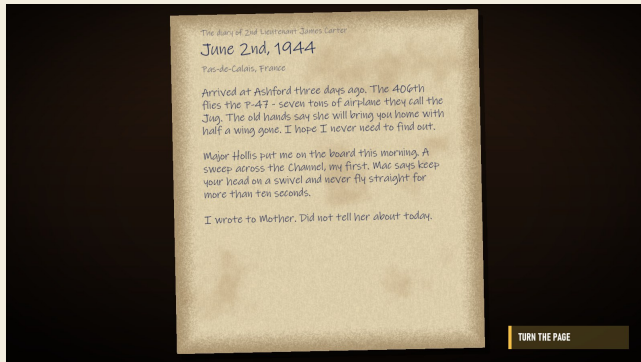
Strafing at tree-top height along a road or an airfield, seen from an isometric camera.

- Slide left and right, climb and dive. Your rounds strike ahead of you, and **the lower you fly the tighter the pattern.**
- Low is also where the poplars, barns and church towers are. Hit one and you may not get a second chance.
- Your shadow on the ground is the best guide to your height.
- Trucks burn easily; tanks shrug off most of what you throw at them.
- **Six variants:** convoy, airfield, train busting, a coastal battery among bunkers and surf, barges on the Seine, and a headquarters in a village.

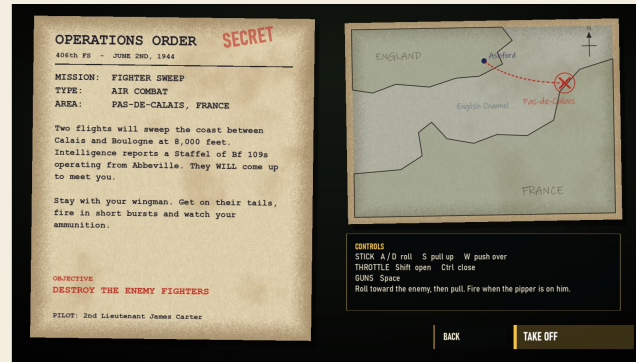
IN THE ORIGINAL

Wings flew its strafing runs diagonally across an isometric landscape, with height as the second axis. That is unchanged.

The career



The diary sets the scene before each mission.



The briefing: orders, route map and controls.



The combat report: results, promotion, medals.

Your pilot. You name him, say where he is from and spend points on four aptitudes. A successful sortie raises the aptitude it exercised.

Flying	How tightly the aircraft turns and how quickly it answers.
Shooting	Tighter bursts and harder hits.
Mechanical	Less damage taken; more ammunition and bombs carried.
Stamina	The odds of surviving when you are shot down.

Rank and decorations. Score earns promotion from 2nd Lieutenant up to Lieutenant Colonel. Five decorations can be won: the Air Medal, the Distinguished Flying Cross for five aerial victories, the Silver Star for an outstanding sortie, the Purple Heart and the Distinguished Unit Citation.

Going down. Being shot down does not always end a career. Depending on his stamina a pilot may bail out and walk home, come back wounded, or be killed. A dead pilot is remembered on the squadron roster, and a replacement you create takes over the campaign where it stands.

Progress is saved after every sortie.

Seven weeks over Normandy

Eighteen missions from the eve of the invasion to the Seine, each a different variant. Any of the three mini-games can also be started from the menu as **Instant Action**, which picks a random variant, time of day and weather and does not touch your career.

DATE	MISSION	TYPE	VARIANT	WHERE	DIFF.
June 2	Fighter Sweep	Air combat	Sweep	Pas-de-Calais	
June 4	The Rail Yard	Bombing	Rail yard	Amiens	
June 6	D-Day	Ground attack	Convoy	Road to Caen	
June 7	Little Friends	Air combat	Bomber escort	Caen	
June 9	The Battery	Ground attack	Coastal battery	Utah Beach	
June 10	Bandits over the Beachhead	Air combat	Sweep	Sainte-Mere-Eglise	
June 12	Train Busting	Ground attack	Train	Lisieux	
June 14	The Bridge	Bombing	Bridge	River Orne	
June 16	Bombers Inbound	Air combat	Intercept	The anchorage	

DATE	MISSION	TYPE	VARIANT	WHERE	DIFF.
June 18	Airfield Attack	Ground attack	Airfield	Evreux	
June 20	Noball	Bombing	Launch site	Saint-Omer	
June 23	Diver Patrol	Air combat	V-1 patrol	Kent coast	
June 26	The Harbour	Bombing	Harbour	Le Havre	
June 30	Jumped	Air combat	Ambush	Falaise	
July 4	The Headquarters	Ground attack	Village HQ	Villers-Bocage	
July 8	Column on the Move	Bombing	Moving column	Saint-Lo	
July 12	The Red-Nosed 109	Air combat	Duel	Caen	
July 18	River Traffic	Ground attack	River barges	The Seine	

Controls and how to run it

ACTION	KEYBOARD	GAMEPAD
Steer	W A S D or arrow keys	Left stick
Guns and bombs	Space	A, right trigger
Throttle (air combat)	Shift to open, Ctrl to close	Shoulder buttons
Pause	Esc	Start
Fullscreen	F11	

- **Air combat:** S pulls the nose up, W pushes it down, A and D roll.
- **Bombing:** steer in all four directions over the scrolling ground.
- **Ground attack:** A and D slide sideways, W dives, S climbs.

The exported build

A Windows build is in the export folder of the project. Keep the two files together; the game will not start without the DLL beside it.

```
export\WW2Wings.exe  
export\ww2wings.dll
```

From source

Requires Windows, Godot 4.7, Visual Studio 2022 with the C++ workload and CMake.

```
git submodule update --init  
build.bat  
run.bat
```

The first build compiles the Godot C++ bindings and takes a few minutes.

Everything is made in code

The game is written in C++ as a Godot 4.7 GDExtension, in roughly 10,900 lines. The Godot project itself is a single scene with a single node. The repository contains no models, textures or audio.

- **Aircraft** are lofted from cross-sections and aerofoil profiles. Paint, camouflage, invasion stripes, national markings and panel lines are drawn by a shader.
- **The landscape** is a shader too: the irregular fields and hedgerows of Normandy, rivers, roads, railways and runways.
- **Sound** is synthesised when the game starts: engines, guns, explosions, the falling-bomb whistle and the music.
- **Lighting** uses Godot's Forward+ renderer with a procedural sky, shadows, ambient occlusion, fog and bloom.



The model viewer: every aircraft, vehicle and building is generated at start-up.

STATUS OF THE VERTICAL SLICE

- The full loop plays from menu to ending, and has been run unattended from start to finish.
- Player controls and the pause menu were checked with simulated key presses.
- Difficulty has only been tuned against a simple autopilot. **It has had no structured play-testing.**
- Sound was checked numerically, not by ear.